Total no. of Pages: 2 Roll no. ....

Degree: B.Tech. Semester: 3 Course work

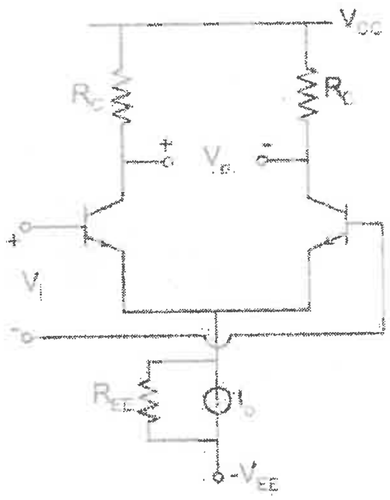
MID-SEMESTER EXAMINATION, June 2026

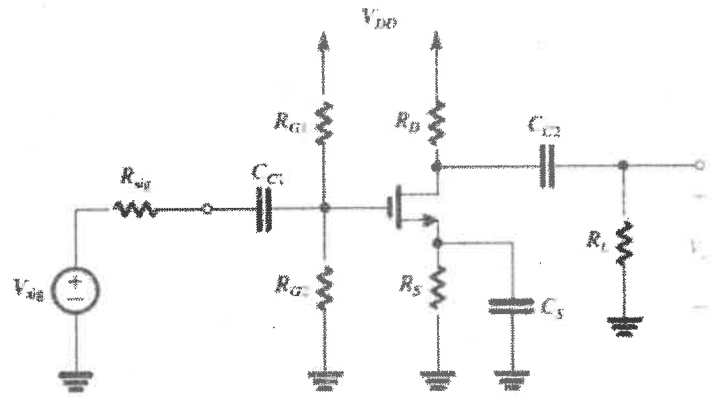
Course Title: Microelectronics Circuits and Application

Course Code: ECECC304/EIECC07

Duration: 1:30 Hours Max. Marks: 20/15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	For the differential amplifier circuit shown in Fig. 1(a), given that $V_{CC} = 12\,V, V_{EE} = -12\,V, I_o = 400\,\mu A, \beta = 100, R_{EE} = 270\,k\Omega, R_C = 47\,k\Omega$, and $V_A = \infty$: **(i)** Determine the differential-mode voltage gain (A_d). **(ii)** Determine the common-mode voltage gain (A_{cm}).  Fig. 1	3/2	CO1
1b	Draw the small circuit equivalent circuit of the differential amplifier shown in Fig.1	1	CO1
2a	Draw the circuit of a MOSFET cascode amplifier and explain why it exhibits higher output resistance than a common-source amplifier.	3/2	CO1
2b	A differential amplifier has differential gain $A_d = 250$and common-mode gain $A_c = 0.5$. Determine the CMRR in dB.	1	CO1

3a	In a simple MOS current mirror, transistor M_1 has $(W/L)_1 = 10$ and carries a reference current of $100\ \mu A$. If $(W/L)_2 = 30$, determine the output current. State one limitation of a simple current mirror.	3/2	CO2
3b	State Miller's Theorem and provide a detailed proof. Derive the expressions for the Miller equivalent impedances at the input and output terminals.	1	CO2
4a	A cascode amplifier has $g_m = 5\ mS$ and load resistance $R_D = 20\ k\Omega$. Determine the approximate voltage gain. State one advantage of a cascode configuration.	3/2	CO2
4b	Draw the high-frequency hybrid-π model of a BJT. Briefly explain the significance of C_π and C_μ.	1	
5a	For the common-source MOSFET amplifier shown in Fig. 2, derive an expression for the overall lower 3-dB cut-off frequency (f_L) using the Short-Circuit Time Constant (SCTC) method. Clearly state all assumptions made during the derivation.  Fig. 2	3/2	CO3
5b	Draw the small circuit equivalent circuit of the differential amplifier shown in Fig. 2.	1	CO3